

SAI TEJA SOMU

@ somuteja@gmail.com +91 8897066870 Bengaluru,India

WORKING EXPERIENCE

Senior AI Engineer

GoSats

April 2025 - October 2025 Bengaluru, India

- Built a multi-agent shopping assistant using LangChain, OpenAI, and Groq APIs with hybrid RAG over Qdrant. Optimized hybrid search with FastEmbed, reducing P50 latency from 2.5s to 0.12s.
- Converted the multi-agent shopping assistant into a Voice agent using Sarvam's Saaras model for audio transcription of audio input of the user which were suited for Indian accents and languages. More than a dozen speech to text services were evaluated on a sample consisting of audios with diverse Indian accents and languages, evaluated the services on word error rate and custom metrics. Used elevenlabs-v2.5-flash model for text to speech after using voice cloning.
- Optimized query handling with faster inference services, async I/O, and multi-threaded parallelism, reducing median latency from 12s to 5s.
- Productionized a support agent with APIs for ticketing, transaction lookup, and card status, using Vertex AI for inference, Docker for containerization, and Cloud Run for deployment.
- Established robust evaluation pipelines using synthetically generated edge case data to benchmark agent reliability and guide iterative improvements.

Machine Learning Engineer

Stealth Mode Startup

May 2024 - March 2025 Bengaluru, India

- Designed an automated documentation generation system using GraphRAG, integrating data from Slack, Google Docs, CRM platforms, and public sources into a knowledge graph. Built pipelines that automatically detected emerging topics and generated structured documentation; Supabase (Postgres) and FastAPI were used for storage and data APIs. Data ingestion involved dealing with data of different modalities like videos, audios, text files etc. Used Assembly AI speech to text service for dealing with audio and video files. Llamaindex framework was used for RAG pipelines and also for extracting texts from unstructured file formats like .docx, .pdf files.
- Developed LLM-based models to evaluate customer service agents' technical knowledge, problem-solving skills, and empathy. Leveraged Assembly AI for speech-to-text transcription and built workflows to automatically surface probable bugs and feature requests from customer conversations.

Machine Learning Engineer

JustAnswer India

July 2022-May 2024 Bengaluru, India

- Trained and deployed fine-tuned GPT-3.5 models for an end-to-end question categorizer, achieving 96%+ accuracy and cut miscategorization by 40%. Each miscategorization costs 8 minutes of manual effort and hurts user experience. Multiple methods were explored for categorization: such as, KNN using embeddings of chat conversations, RAG in which the LLM predicts the category after feeding relevant examples in the few-shot prompt to LLM etc.

ACHIEVEMENTS

- All India Rank 1 in GATE Mathematics 2017**
National topper (Rank 1) in GATE Mathematics.
- All India Rank 2 in CSIR UGC NET Mathematical Sciences**
Rank 2 in CSIR UGC NET Mathematical Sciences, Dec 2016.

STRENGTHS

Data Science

LLMs

Retrieval-Augmented Generation (RAG)

Prompt Engineering

Natural Language Processing

Machine Learning

Statistical Analysis

Python

SQL

Research

PROGRAMMING LANGUAGES

Python

SQL

C++

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LANGUAGES

English

Hindi

Telugu

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EDUCATION

Integrated Master of Science in Applied Mathematics
Indian Institute of Technology, Roorkee (IITR)
2011 – 16
Cumulative GPA 8.04/10

- Clustered 2M+ SEO pages using OpenAI embeddings with K-Means, consolidating content at scale. This initiative improved site structure and drove a 50% increase in organic traffic.
- Worked on LTV prediction project where I have trained linear regression, gradient boosting models like XgBoost, LGBM. Decreased the MAE by 5%.
- Built an agentic workflow for generating ad descriptions and ad headlines based on targeted keywords that aligns with business leveraging Claude, GPT, and miscellaneous LLMs.

Data Scientist

WhamTech India Dedicated Development Center

📅 May 2018-July 2022

📍 Hyderabad, India

- Developed decision-tree-based models to automatically identify sensitive attributes in enterprise databases, enabling faster data-privacy audits and ensuring compliance with regulations such as GDPR and HIPAA.
- Used the GATE NLP framework to extract entities such as policy numbers, phone numbers, organization names, and other domain-specific fields from insurance documents, making the data available for further analysis.
- Implemented the Expectation-Maximization algorithm from scratch in C++ to estimate Fellegi-Sunter m and u probabilities for probabilistic record linkage, enabling unsupervised entity resolution across large databases.
- Built C++ libraries integrated with SQL/Postgres to perform probabilistic record linkage, using Naive Bayes and EM to identify and merge duplicate records for Master Data Management.

PUBLICATIONS

1. Sai Teja Somu and Pandey R K, On unavoidable obstructions in Gaussian walks, INTEGERS Electronic Journal of Combinatorial Number Theory, Volume 16(2016), A87, Pages 1-13
2. Sai Teja Somu, On the coefficients of divisors of $x^n - 1$, Journal of Number Theory, Volume 167, October 2016, Pages 284-293
3. Sai Teja Somu, On the distribution of numbers related to the divisors of $x^n - 1$, Journal of Number Theory, Volume 170, January 2017, Pages 3-9
4. Sai Teja Somu, On integer sequences in product sets, Journal of Number Theory, Volume 180, November 2017, Pages 756-768
5. Sai Teja Somu, Faisant A., Grekos G., Pandey R.K., Additive Complements for a given Asymptotic Density, Mediterr. J. Math., 18(1), paper No 25, 1-13, 2021
6. Sai Teja Somu, Kukla A., On Some Generalizations to Floor Function Identities of Ramanujan, INTEGERS Electronic Journal of Combinatorial Number Theory, Volume 22(2022), A33, Pages 1-11
7. Sai Teja Somu, Grekos, G., Pandey, R.K., Sumsets with Prescribed Lower and Upper Asymptotic Densities. Mediterr. J. Math. 19, 201 (2022).
8. Sai Teja Somu, Mishra V., On power values of sum of divisors function in arithmetic progressions, Indian J Pure Appl Math (2023).
9. Sai Teja Somu, Johnathan Haw, Vincent Nguyen, Duc Van Khanh Tran, On some series with gaps, Journal of Mathematical Analysis and Applications, Volume 528, Issue 1, 2023, 127479, ISSN 0022-247X, <https://doi.org/10.1016/j.jmaa.2023.127479>.
10. Sai Teja Somu, Mishra V., On a generalization of Tupper's formula for m colors and n dimensions, Discrete Mathematics, Volume 346, Issue 12, 2023, 113600, ISSN 0012-365X, <https://doi.org/10.1016/j.disc.2023.113600>.
11. Sai Teja Somu, Li T.H.S, Kukla A., On Some Results On Practical Numbers, INTEGERS Electronic Journal of Combinatorial Number Theory, Volume 23(2023), A68, Pages 1-9.
12. Sai Teja Somu, Tran D.V.K., On Sums of Practical Numbers and Polygonal Numbers, Journal of Integer Sequences, Volume 27(2024), Article 24.5.1, Pages 1-12.
13. Sai Teja Somu, Grekos, G., Pandey, R.K., Sumsets with Prescribed Lower and Upper Asymptotic Densities II, Uniform Distribution Theory, 2024, 19 (1).